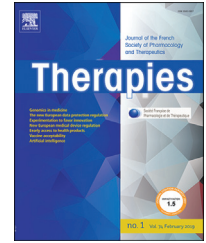




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PHARMACOLOGY AND ARTIFICIAL INTELLIGENCE

Artificial intelligence in personalized prescription: A narrative review of promise, peril, and practicality

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Summary The integration of artificial intelligence (AI), particularly large language models (LLMs), into clinical prescription practices represents a transformative shift in healthcare, promising to offer enhanced therapeutic precision, reduced errors, and personalized care. However, rapid adoption is outpacing validation and governance. This narrative review critically examines the benefits, risks, and implementation challenges of AI in pharmacology, with an emphasis on prescribing use cases, highlighting the rapid adoption driven by physician constrained time and increasing complexity of medical knowledge, yet outpaced by validation frameworks. Key benefits include optimized decision support, as suggested by real-world systems like the Penda Health AI Copilot. However, inherent risks such as hallucinations, lack of explainability, spending inflation, and clinical de-skilling pose significant threats, potentially eroding trust and patient safety. We contrast regulatory philosophies [European Union (EU) precaution under the medical devices regulation (MDR) and AI Act vs U.S. flexibility via Food and Drug predetermined change control plans (FDA PCCP)] and spotlight a persistent grey zone in which general-purpose LLMs are used clinically without medical-device oversight. To operationalize trustworthy use, we foreground rigorous evaluation and outline an “architecture of trustworthy clinical AI” (retrieval-augmented generation, structured clinical prompting with abstention/uncertainty, and hybrid LLM–rule safety layers). Finally, we propose an outcomes-first paradigm adapted from French health technologies assessment (HTA) — service rendered by AI (SR-AI) and improvement of service rendered by AI (ISR-AI), with patient

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outcomes and system resilience as primary endpoints — and recommend a default stance that any generative AI used in health be treated as a medical device, with general-purpose LLMs permitted only via certified clinical wrappers that close the MDR-inconsistent gap.

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Abbreviations

AI	artificial intelligence
AMA	American Medical Association
CDSS	clinical decision support system
CHART	Chatbot assessment reporting tool
EU	European Union
FDA	Food and Drug Administration
HAS	<i>Haute autorité de santé</i>
HTA	health technologies assessment
ISR	improvement of service rendered
LLM	large language model
MDR	medical devices regulation
NICE	National Institute for Health and Care Excellence
PCCP	predetermined change control plans
RAG	retrieval-augmented generation
SaMD	software as a medical device
SR	service rendered
XAI	explainable AI

Introduction

The advent of artificial intelligence (AI) in medicine is arguably one of the most significant technological developments of the early 21st century. More specifically, over the three years, the integration of large language models (LLMs) into clinical practice has reached what appears to be an irreversible tipping point, fundamentally transforming the landscape of pharmacology and prescription practices [1].

Previous generations of clinical decision support system (CDSSs) algorithms were often characterised by rigid interfaces and rule-based architectures. While they excelled at applying predefined rules, they struggled with the ambiguity and complexity inherent in clinical medicine [2]. In contrast, LLMs offer intuitive, conversational interfaces and the ability to reason across multiple, disparate data domains simultaneously — from patient demographics and unstructured clinical notes to biological data and the latest clinical guidelines [3,4], thereby enabling the suggestion of a prescription protocol that would have been unachievable through prior expert systems [2]. This unique capacity to understand, summarise, and generate human-like text place LLMs not merely as a tool for information retrieval, but as an active partner in the clinical reasoning process itself [5].

In this regards, AI in pharmacology represents a qualitative shift rather than an incremental improvement [6]. Its

adoption in clinical practice has accelerated dramatically, driven by converging factors that make this technological integration both inevitable and necessary [7]. Recent evidence underscores this leap: a 2024 American Medical Association (AMA) survey revealed that nearly two-thirds (66 %) of physicians now utilize AI technologies in their practice, marking a 78 % increase from just 38 % in the previous year [8,9].

This rapid uptake was not primarily motivated by a top-down strategic push for clinical innovation, but rather by a bottom-up, pragmatic response to the systemic crisis of physician constrained time and increasing complexity of medical knowledge [10,11]. The first catalyst for AI adoption has indeed been the urgent need to address the administrative burdens that contribute to physician stress and detract from patient care [12]. The same AMA survey found that 57 % of doctors see automating paperwork as the top opportunity for AI to combat burnout [13]. The top uses for AI among physicians were administrative, including documentation of billing codes and medical charts, creation of discharge instructions, and generation of draft responses to patient portal messages. Ambient intelligence tools, such as AI-powered digital scribes that transcribe patient encounters and generate clinical notes, have also flourished, demonstrating significant potential in reducing this cognitive load [14]. One large primary care network, for example, reported a 54 % drop in documentation-related cognitive load after implementing an ambient AI scribe [15] and other studies have shown that such tools can save physicians minutes to hours per day, leading to boosts in job satisfaction and a reduction in “pajama time”, the after-hours documentation burden [16].

This initial adoption for administrative tasks, however, appears to be the first stage of a more profound transformation. The trajectory of adoption reveals a critical dynamic: as physicians become comfortable with AI for lower-risk administrative functions, their appetite for and acceptance of clinical AI applications increases substantially. This “familiarity effect” creates a clear pathway toward the integration of AI into more sophisticated and higher-risk tasks, including personalised prescribing and clinical decision support [5].

This creates an adoption–validation gap: the pragmatic spread of AI outpaces the rigorous evaluation needed for high-stakes clinical scenarios [17,18]. While a tool used to draft a billing or visit summary carries low risk, the same underlying technology used to recommend a ther-

apy regimen carries higher risk [19]. This narrative review aims to critically evaluate the benefits, risks, and practical implementation challenges of AI in prescription practices, arguing for a responsible deployment framework that moves beyond the current fascination with scenarios-based studies on general-purpose models toward specialised, clinically validated tools [20].

The benefit–risk equation for AI in prescription

The integration of AI into the process of prescribing medication presents a classic benefit–risk equation. On one side, the potential for AI to enhance therapeutic precision, reduce errors, and personalize care is substantial and increasingly well-documented [4,21]. On the other, the introduction of this powerful yet fallible technology into a high-stakes clinical domain introduces a new class of risks that demand careful management [4,22].

Quantifiable benefits in therapeutic optimisation

LLMs show considerable potential in providing clinical decision support by suggesting treatment recommendations, and retrieving relevant medical knowledge on demand [23]. While still an emerging field, preliminary studies have shown that LLMs can process extensive medical literature and patient data swiftly, providing therapy recommendations across multiple clinical specialties and assisting with complex tasks where up-to-date knowledge is critical [22]. Although most evaluations have focused on the technical performance of algorithms, there is a clear trajectory toward clinical implementation.

Notably, a systematic review of 761 studies on the topic found an exponential increase in research output, underscoring the expanding capabilities and clinical potential of LLMs [22]. However, the same review noted that evaluations have predominantly focused on general-domain models like ChatGPT (93.55 % of cases) and mostly in simulated scenarios, highlighting the need for more rigorous, standardized frameworks to validate their performance and safety in real-world clinical workflows with dedicated tools.

While algorithm-specific case studies are valuable, evidence from comprehensive, real-world clinical implementation is also critical. The recently published Penda Health AI Clinical Copilot, in partnership with OpenAI, has provided a landmark example of a well-designed LLM-based system functioning in a general practice setting in Kenya. In this large-scale, controlled study involving 5666 patient encounters, clinical officers using the AI copilot demonstrated a 13 % relative reduction in treatment errors compared to a control group [24], as rated by blinded independent physicians that aimed to identify clinical errors. The success of the Penda system was not attributed solely to the power of the underlying model but to three key factors: a capable model, a clinically-aligned implementation, and active deployment strategies. The AI “copilot” listened to the patient encounter via ambient speech recognition and provided real-time safety prompts and alerts to the clin-

ician, who retained the final choice of therapy. The tool was co-developed with healthcare professionals to ensure it supported their workflow, providing tiered “traffic-light” alerts to minimise cognitive burden. This study exemplifies the augmentation-rather-than-replacement paradigm, demonstrating that a properly integrated AI assistant can serve as a powerful clinical safety net that improves care quality before errors reach the patient. Of note, this study had limits in generalizability: the study was conducted with clinical officers — practitioners who complete three years of academic training plus a one-year supervised internship — rather than physicians. Results should therefore be interpreted cautiously across other health-system contexts, even as the study illustrates the potential of a well-designed AI CDSS for less trained professionals.

Another study by Amazon and published in *Nature Medicine* [25] showed that LLMs can substantially reduce medication direction errors in online pharmacies, where ambiguous or incomplete prescription instructions are a frequent source of patient harm. By automatically standardizing directions and flagging inconsistencies before dispensing, AI systems reduced error propagation across the prescribing–dispensing continuum, reinforcing their role as active safeguards in pharmacotherapy.

However, a 2025 scoping review on generative AI for preventing medication-related harm highlighted that, although some results are promising for the integration of such tools CDSSs, the overall lack of rigorous evaluations warrants caution and underscores the need for robust evidence of real-world effectiveness [26].

A critical examination of inherent risks

The deployment of AI in prescription practices indeed introduces significant and novel risks that must be proactively managed [19]. These risks are not merely technical but are deeply intertwined with clinical practice, economics, and professional identity.

Hallucinations and explainability

One of the most concerning risks associated with LLMs is their propensity for “hallucination” — the generation of information that is nonsensical, factually incorrect, or entirely fabricated, yet presented with the same confident and fluent tone as factual information [27,28]. This phenomenon is not a bug but an inherent characteristic of the underlying technology. Most LLMs are auto-regressive, meaning their core function is to predict the next most probable word in a sequence based on patterns learned from a massive training dataset; their objective is linguistic coherence, not factual accuracy [3].

This technical origin has profound implications for clinical practice, including unsafe dose suggestions, fabricated citations, missed or false contraindications, and misdiagnosis-driven prescriptions. The lack of an inherent “truth verification” mechanism means that without specific safeguards, LLMs cannot be trusted in high-stakes environments like prescribing. The risk is compounded by automation bias, where a busy clinician might uncritically accept a plausible-sounding but false AI recommendation [29,30].

Closely related to hallucination is the “black box” nature of many advanced AI models [31]. When an AI system recommends a specific medication, clinicians and patients have a right to understand the reasoning behind that recommendation. Without this transparency, clinical accountability is eroded, and professional responsibility becomes impossible to maintain.

Explainable AI (XAI) is a critical field aimed at solving this problem [31]. XAI encompasses techniques designed to render AI decision-making processes transparent and interpretable. These include feature attribution methods that identify which patient factors most influenced an output, and attention mechanisms that can point to a specific lab result or comorbidity as the reason for a dose adjustment. By integrating XAI, or combining AI with traditional, rule-based systems can move from opaque pronouncements to transparent suggestions, fostering the trust necessary for true human-AI collaboration.

The economic paradox: cost reduction vs. expenditure inflation

AI is often touted for its potential to generate massive healthcare savings, with some estimates suggesting it could reduce US healthcare spending by 5 to 10 percent (\$200 billion to \$360 billion annually) [7].

However, a critical counter-risk is the potential for AI to inadvertently *increase* prescription spending if not designed with economic principles in mind. An AI system optimised solely for clinical efficacy, without considering cost, might consistently recommend newer, more expensive brand-name medications over equally effective and affordable generic alternatives. If an AI is trained on drug catalogs or promotion-heavy literature from industries known to influence prescribing habits, it might unknowingly perpetuate that bias. The previously cited Amazon study underscores large technology firms’ interest in this space; because Amazon also operates as a U.S. pharmacy provider, this dual role raises potential conflict-of-interest concerns. To prevent AI from becoming a driver of unsustainable healthcare expenditure, regulatory and HTA bodies must evaluate AI tools not only for clinical performance but also for cost-effectiveness.

The specter of clinical de-skilling

Perhaps the most insidious long-term risk of AI integration is the potential for “de-skilling” — the erosion of clinicians’ fundamental knowledge and reasoning abilities due to over-reliance on automated systems [10]. Observers warn that excessive automation “may diminish clinical skills,” since hands-on reasoning is needed to maintain expertise. A qualitative survey of UK clinicians found that while experienced consultants used AI as a tool to be critically evaluated, they worried that less experienced doctors were prone to over-relying on its output. A physician noted, “You have to have the experience to be able to correct it. If you haven’t done thousands of patients, you cannot correct the output”. This suggests that over-reliance could prevent junior clinicians from ever acquiring the deep, tacit knowledge that comes from hands-on experience. This could lead to a “brittle” healthcare system, where practitioners become dangerously ill-equipped to function when AI systems fail or encounter rare cases outside their training data. The interplay between

these risks creates a negative feedback loop: the “black box” nature of a tool prevents learning, which accelerates de-skilling, which in turn lowers the defenses against AI hallucinations [31].

Navigating the evolving regulatory landscape

The rapid emergence of AI in healthcare has created significant challenges for regulatory bodies worldwide, leading to fragmented and evolving global frameworks.

Divergent philosophies: EU vs. US

A comparison between Europe and the United States reveals two distinct approaches to governing AI in medicine.

The European Union has established a comprehensive and precautionary framework anchored by the medical device regulation (MDR) and the landmark AI Act [32]. Under the MDR, most AI prescription tools qualify as class IIa or higher software as a medical device (SaMD), mandating rigorous assessment by an independent “notified body” to demonstrate safety and performance. The AI Act complements this by imposing additional obligations on such “high-risk” systems, including strict data governance, detailed documentation, and provisions for meaningful human oversight. Non-EU-member countries, such as the UK or Switzerland, have largely adopted analogous approaches. Beyond initial regulatory clearance, achieving reimbursement — which facilitates quicker market entry in these centralized healthcare systems — typically necessitates assessment by national health technology assessment (HTA) bodies, like France’s Haute Autorité de santé (HAS) or the UK’s National Institute for Health and Care Excellence (NICE). These entities rigorously examine clinical effectiveness and incremental value, creating a dual-layer safeguard for both safety and societal benefit. That said, medical devices intended for professional use have traditionally escaped direct HTA evaluation and reimbursement in Europe, including France, as they were regarded merely as adjuncts to medical procedures, with funding tied to the acts themselves rather than the devices. Yet, as AI progressively assumes roles in human decision-making or procedural tasks, this paradigm is increasingly inadequate and lacks a well-established regulatory structure. Recognizing this gap, the HAS is now addressing it in its 2025–2030 strategy by formulating a tailored HTA framework specifically for professional medical devices [33].

In contrast, the United States Food and Drug Administration (FDA) has pursued a more flexible and adaptive regulatory approach where software used as SaMD are regulated drawing on international principles; products are reviewed via existing device pathways (510(k), De Novo, or PMA) based on risk and novelty [34]. Moreover, contrary to most European countries, the U.S. lacks a single, centralized HTA authority; reimbursement and coverage decisions are made by a fragmented landscape of public and private payers. This can lead to faster market access for some innovations, but also greater variability in coverage and the adoption of innovations with unproven added value.

The regulatory paradox of general-purpose AI

These frameworks expose a critical and dangerous gap when it comes to general-purpose AI systems like widely accessible chatbots [35]. The central pillar of medical device regulation — both in Europe and the US — is “intended use”. Since developers of tools like ChatGPT do not market them for a specific medical purpose, they fall outside the definition of a medical device, even though they are widely used by patients and clinicians for medical queries. They are therefore not required to undergo the clinical validation or post-market surveillance mandated for SaMD. This creates a profound paradox: a specialized AI algorithm for a narrow task faces years of scrutiny, while a more powerful, general-purpose LLM capable of answering infinite medical questions is available to billions with no clinical oversight. This regulatory loophole provides a perverse incentive for tech companies to keep tools “general purpose” to avoid the medical device regulatory pathway, representing a significant, unaddressed public health risk.

The imperative of rigorous evaluation and reporting

While the promise of AI in personalized prescribing is compelling, its safe translation into clinical workflows requires more than technical performance; it demands rigorous, transparent, and standardized evaluation. Recent experience with LLMs in healthcare underscores this need. Within a year of ChatGPT’s release, more than 130 studies had been published assessing its use for medical advice, yet a systematic review revealed striking heterogeneity and incomplete reporting: few studies detailed their prompting strategies, less than 40 % described query protocols, and raw model outputs were rarely shared [36]. Such opacity prevents independent replication and undermines confidence in clinical validity, creating risks of misinterpretation and patient harm.

To address these shortcomings, the Chatbot Assessment Reporting Tool (CHART) was recently introduced, offering a 12-item, 39 sub-item checklist to guide transparent reporting of LLM studies in health contexts [37]. CHART builds on earlier methodological frameworks developed for AI in medicine, such as CONSORT-AI for randomized trials [38]; SPIRIT-AI for trial protocols [39], TRIPOD-AI and PROBAST-AI for prediction models [40,41], and DECIDE-AI for early-phase evaluations [42]. Collectively, these initiatives emphasize that evaluation of AI tools must move beyond accuracy metrics toward reproducibility, safety profiling, and disclosure of data provenance, algorithmic updates, and ethical safeguards.

In the domain of personalized prescribing, such standards are even more critical. Unlike administrative automation, prescribing carries high clinical stakes, where errors in dosage, contraindications, or drug interactions can have immediate and severe consequences. Evaluation must therefore encompass not only what an AI system can do but how it achieves its outputs, how reproducible these are across populations, and what added value they provide over existing practice. Transparent reporting, pre-registration

of evaluation protocols, and independent validation in real-world clinical settings should be considered minimal requirements.

Establishing robust evaluation frameworks, potentially tailored analogues of CHART for prescription AI, would close the current adoption-validation gap. By mandating methodological rigor, transparency, and reproducibility, such standards can ensure that AI systems deliver trustworthy augmentation of clinical judgment rather than unchecked automation. This evaluation layer forms the necessary bridge between rapid adoption and the value-based frameworks later proposed in this review.

From general-purpose AI to validated clinical tools: a proposed architecture of trustworthy clinical AI

The future of AI in prescription should probably not lie with standalone, general-purpose models, but with specialised, validated clinical tools built upon an architecture of trust.

Building a clinical AI tool that is safe, reliable, and trustworthy requires more than just a powerful LLM; it requires a deliberate system architecture incorporating multiple layers of safety and verification [20].

Advanced clinical prompt engineering

In a clinical context, prompt engineering is a highly specialised discipline for designing inputs to elicit reliable and safe outputs [43]. This involves using structured input templates to ensure all relevant clinical data is provided consistently, building in safety constraints and guardrails (e.g., instructing the AI to never prescribe off-label or to flag ambiguous cases), and designing prompts that encourage the AI to quantify its uncertainty or state when it cannot safely answer. Studies have shown that carefully crafted medical prompts significantly improve accuracy [44].

Retrieval-augmented generation (RAG): grounding AI in verifiable truth

Retrieval-augmented generation (RAG) is arguably the most critical technology for making LLMs suitable for clinical use [45]. A standard LLM generates responses based solely on its static, pre-trained data. RAG changes this by connecting the LLM to an external, dynamic, and authoritative knowledge base in real-time. In a prescription application, the RAG process first retrieves relevant information from trusted sources — such as up-to-date clinical guidelines, drug interaction databases, the local formulary, and the patient’s specific EHR data — *before* the LLM generates a response. The LLM is then instructed to synthesize an answer based only on this provided context. This “grounds” the AI’s response in verifiable sources, dramatically reducing the likelihood of hallucination. Furthermore, a well-designed RAG system can cite its sources, allowing the clinician to easily verify the information. Early reports show RAG-enhanced AI can cut reduce the hallucination rate by 40 % versus more basic systems [46].

Hybrid systems: combining strengths

The most robust implementations probably involve a hybrid approach that combines the strengths of modern LLMs with the reliability of traditional, rule-based expert systems. An integrated system can leverage the best of both: the LLM acts as a sophisticated “front-end” to interpret natural language and unstructured notes, while a traditional rule-based engine applies hard-coded safety checks. This modular architecture, leveraging the nuance of AI with the reliability of classical CDSS rules, represents a more mature and safer paradigm.

A call for a “service rendered” of AI-CDSS and guardrails for general-purpose LLMs

As we stand at the threshold of widespread AI integration, this review reveals both transformative potential and substantial challenges. The benefits are compelling, with systems like the Penda Health AI Copilot proving that properly integrated AI can function as an effective clinical safety net. However, these gains are contingent upon rigorous validation, as the risks of hallucination, opaque reasoning, and clinical de-skilling are equally significant and deeply interconnected.

To ensure that AI serves the best interests of patients, we call for the adoption of a more rigorous and holistic evaluation paradigm for AI based CDSS aimed to improve care quality, as opposed to tools boosting administrative efficiency. Drawing inspiration from the well-established French HTA framework, we propose a structured approach to AI evaluation that adapts the concepts of *service médical rendu* (SMR) and *amélioration du service médical rendu* (ASMR).

Service rendered by AI (SR-AI)

This would assess the absolute clinical benefit of an AI CDSS. It answers the fundamental questions: Does this tool demonstrate efficacy in improving medico-economic or patient-relevant outcomes in rigorous studies? What is its safety profile compared to the standard of care? Does it address a clinical need?

Improvement of service rendered by AI (ISR-AI)

This would evaluate the added value of an AI tool over existing standards of care. It answers the comparative questions: How much better is this tool than current practice? Is it cost-effective? Does it contribute to healthcare system efficiency or improve health equity?

This dual framework forces a shift in perspective from what the technology *can do* to what it *should do* and *improve* to provide meaningful value.

In parallel, we recommend a default regulatory stance whereby any generative AI used in health is treated as a medical device — at minimum Class I — with mandatory quality management, post-market surveillance, and meaningful human oversight. General-purpose LLMs should not be

used for patient-specific medical purposes unless embedded within a certified clinical wrapper that meets these requirements; absent such safeguards, their clinical use should be restricted or prohibited. This regulatory grey area runs counter to the purpose of the EU MDR — to ensure safety, performance, and accountability for software that influences clinical decisions — and should be closed.

To conclude, the future of AI in pharmacology lies not in replacing clinical judgment but in augmenting it with validated, transparent, and value-driven tools that have cleared these high bars. This path demands action from all stakeholders: developers must build for clinical value, clinicians must become critical consumers of AI, and regulators must tie market access to demonstrated benefit. By insisting on the highest standards of validation and safety, we can harness AI's power to personalize therapy while upholding the trust at the heart of medicine.

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